

I/Q Imbalance Pre-Compensation for OFDM Systems Using RF Feedback and BLSTM Predistortion

Jihoon Lee and Jin Kwan Kim

Abstract—In this paper, we propose two novel predistortion schemes to mitigate I/Q imbalance in Orthogonal Frequency Division Multiplexing (OFDM) communication systems: RF feedback-based and deep learning-based schemes. With the RF feedback-based scheme, gain and phase errors are estimated in real time using feedback from the RF signal. With the deep learning-based scheme, I/Q imbalance predistortion is performed using Bidirectional Long Short-Term Memory (BLSTM). The simulation results show that the proposed schemes reduce the Bit Error Rate (BER) in OFDM systems.

Index Terms—I/Q imbalance, predistortion, deep learning, OFDM

I. INTRODUCTION

WITH the evolution from 5G New Radio (NR) to the emerging sixth generation (6G) wireless systems, there is an increasing demand for higher data rates, lower latency, and massive connectivity. Both NR and emerging 6G systems adopt Orthogonal Frequency Division Multiplexing (OFDM) waveforms per recent 3GPP standards [1]–[3].

OFDM offers several advantages, including high spectral efficiency and strong robustness against multi-path fading and Doppler effects [4]. OFDM enables efficient FFT-based signal processing, does not require complex equalizers, and is robust to various channel impairments. Therefore, OFDM remains widely adopted in modern communication systems. However, OFDM systems are highly sensitive to impairments such as frequency offset, high Peak-to-Average Power Ratio (PAPR), Direct Current (DC) offset, I/Q imbalance, and phase noise [5], [6].

Modern wireless systems demand highly integrated, low-cost receivers, making Zero Intermediate Frequency (zero-IF) receivers widely adopted [7]. However, while baseband processing occurs in the digital domain, up/down-conversion is performed in the analog domain, where I/Q imbalance is inevitable due to gain and phase mismatches [8]. As wireless systems move toward higher frequency bands and wider bandwidths in 6G, the impact of I/Q imbalance becomes increasingly severe due to heightened sensitivity of analog components to gain and phase mismatches at millimeter-wave frequencies.

Several research efforts have addressed I/Q imbalance [9]–[12]. However, [9]–[11] require additional pilot signals, reducing spectral efficiency, and focus on post-compensation at

the receiver. Since post-compensation operates on the already-distorted received signal, it inherently results in larger residual errors, particularly under non-linear I/Q imbalance. Pre-compensation at the transmitter directly eliminates distortion before transmission, offering inherently better performance. Reference [12] addresses transmitter-side I/Q imbalance by exploiting the mirror interference structure, where paired subcarriers are used to pre-cancel the distortion at the transmitter. However, this method assumes that the I/Q imbalance parameters are perfectly known a priori, which is difficult to guarantee in practice. When the compensation parameters deviate from the true values, the mirror interference is incompletely canceled and manifests as residual self-interference, causing severe performance degradation with increasing parameter mismatch.

The main contributions of this paper are summarized as follows:

- 1) We propose an RF feedback-based predistortion scheme that estimates I/Q imbalance parameters in real time using internal RF feedback, without requiring pilot signals or a priori parameter knowledge.
- 2) We propose a BLSTM-based predistortion scheme that learns the compensation mapping directly from training data, without requiring pilot signals or internal feedback. Unlike conventional linear methods, it can handle non-linear I/Q imbalance characteristics.

The simulation results demonstrate that the proposed predistortion schemes improve the Bit Error Rate (BER) performance of the OFDM system under I/Q imbalance conditions.

II. SYSTEM MODEL

A. OFDM System

In an OFDM system, a high-rate data stream is divided into multiple low-rate data streams. The time-domain OFDM signal $x(t)$ is [4]

$$x(t) = \frac{1}{N} \sum_{n=0}^{N-1} X_n e^{j2\pi n \Delta f t}, \quad 0 \leq t < T \quad (1)$$

where N denotes the number of subcarriers, X_n represents the data symbol on the n th subcarrier, $\Delta f = 1/T$ is the subcarrier spacing and T is OFDM symbol duration. In (1), the time-domain OFDM signals $x(t)$ can be expressed as a sum of in-phase signal $I(t)$ and quadrature signal $Q(t)$ ($x(t) = I(t) + jQ(t)$).

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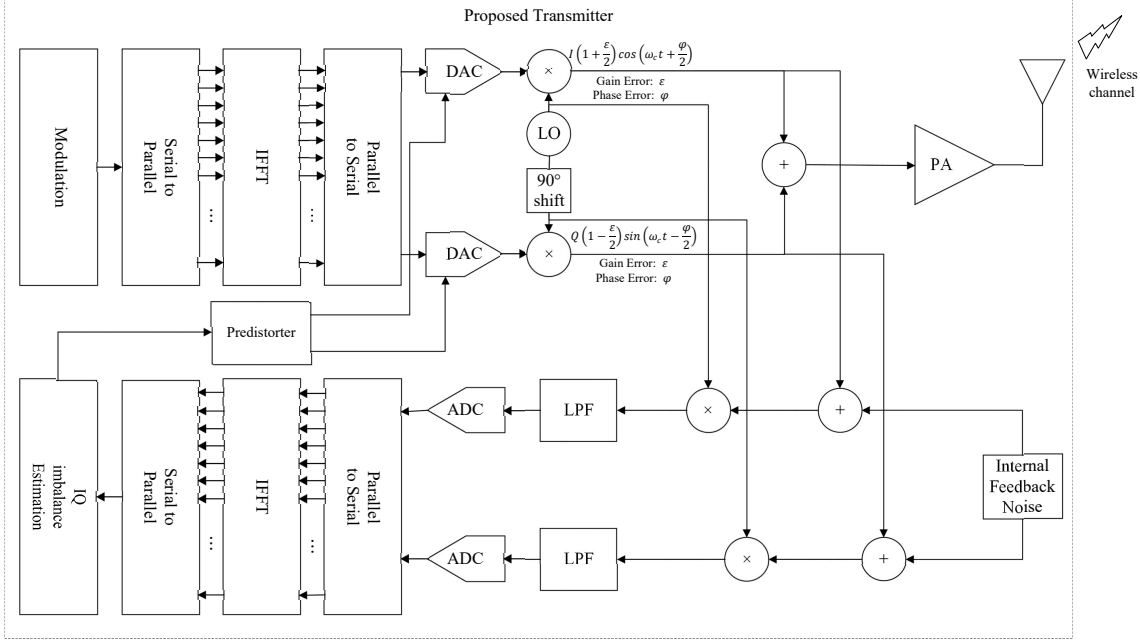


Fig. 1. Block diagram of the proposed transmitter with RF feedback-based I/Q imbalance predistorter.

B. I/Q Imbalance

In the up-conversion process, in-phase and quadrature signals are mapped onto cosine and sine waves of Local Oscillator (LO), respectively. In ideal up-conversion, the RF signal $s(t)$ is

$$s(t) = I(t) \cdot \cos(\omega_c t) + Q(t) \cdot \sin(\omega_c t), \quad (2)$$

where ω_c is the carrier frequency of the RF signal.

The signal with I/Q imbalance $\hat{s}(t)$ can be expressed as [8]

$$\begin{aligned} \hat{s}(t) = & I(t) \left(1 + \frac{\varepsilon}{2}\right) \cos\left(\omega_c t + \frac{\varphi}{2}\right) \\ & + Q(t) \left(1 - \frac{\varepsilon}{2}\right) \sin\left(\omega_c t - \frac{\varphi}{2}\right), \end{aligned} \quad (3)$$

where ε and φ are the gain and phase errors of the transmitter LO, respectively. This model assumes a frequency-independent (flat) I/Q imbalance, which is commonly used for narrowband OFDM systems. In this model, ε represents the amplitude mismatch between the I and Q branches, while φ denotes the phase deviation from the ideal 90° phase difference between the LO signals. While ε and φ are modeled as static parameters in the analytical derivation, in practice these parameters may exhibit non-linear characteristics and practical parameter variations due to temperature variations, component aging, and signal-dependent effects. In this study, ε and φ are used to model only the transmitter-side I/Q imbalance.

III. I/Q IMBALANCE PREDISTORTION SCHEMES

A. RF feedback-based predistortion of I/Q imbalance

Fig. 1 shows the block diagram of the proposed transmitter with an I/Q imbalance predistorter based on the RF signal. For clarity, the wireless channel and receiver are not shown, because the proposed scheme operates entirely at the transmitter side. The internal feedback path samples the Power Amplifier (PA) output and performs down-conversion using the same LO employed for up-conversion within the same transmitter hardware. Consequently, there is no LO mismatch between the up- and down-conversion paths. This differs fundamentally from receiver-side compensation schemes [9]–[11], which require synchronization between separate transmitter and receiver LOs.

The down-conversion signal $\tilde{x}(t)$ can be expressed as the sum of in-phase signal $\tilde{I}(t)$ and quadrature $\tilde{Q}(t)$ ($\tilde{x}(t) = \tilde{I}(t) + j\tilde{Q}(t)$). Then, $\tilde{I}(t)$ and $\tilde{Q}(t)$ can be expressed as

$$\begin{aligned} \tilde{I}(t) = & \int_0^t \hat{s}(\tau) \left(1 + \frac{\varepsilon}{2}\right) \cos\left(\omega_c \tau + \frac{\varphi}{2}\right) + n(\tau) d\tau \\ = & I(t) \left(1 + \frac{\varepsilon}{2}\right)^2 - Q(t) \left(1 - \frac{\varepsilon^2}{4}\right) \sin\varphi + n(t), \end{aligned} \quad (4)$$

$$\begin{aligned} \tilde{Q}(t) = & \int_0^t \hat{s}(\tau) \left(1 - \frac{\varepsilon}{2}\right) \sin\left(\omega_c \tau - \frac{\varphi}{2}\right) + n(\tau) d\tau \\ = & Q(t) \left(1 - \frac{\varepsilon}{2}\right)^2 - I(t) \left(1 - \frac{\varepsilon^2}{4}\right) \sin\varphi + n(t), \end{aligned} \quad (5)$$

where $n(t)$ represent internal thermal noise.

To simplify the analytical derivation, we assume $\varepsilon^2 \ll 1$, so that (4) and (5) can be approximated by (6) and (7).

$$\tilde{I}(t) \approx I(t) + \varepsilon I(t) - Q(t) \sin \varphi \quad (6)$$

$$\tilde{Q}(t) \approx Q(t) - \varepsilon Q(t) - I(t) \sin \varphi \quad (7)$$

While this approximation is strictly valid for small ε , the proposed estimation method is evaluated over a broader practical range ($\varepsilon = 0.3 \sim 0.6$) in Section IV, where the approximation error remains sufficiently small for practical estimation purposes, as validated by the simulation results in Section IV.

Multiply both sides of (6) by I and that of (7) by Q as

$$\tilde{I}(t)I(t) \approx I^2(t) + \varepsilon I^2(t) - I(t)Q(t) \sin \varphi \quad (8)$$

$$\tilde{Q}(t)Q(t) \approx Q^2(t) - \varepsilon Q^2(t) - I(t)Q(t) \sin \varphi \quad (9)$$

Subtracting (9) from (8) yields:

$$\begin{aligned} \tilde{I}(t)I(t) - \tilde{Q}(t)Q(t) \\ \approx I^2(t) - Q^2(t) + \varepsilon (I^2(t) + Q^2(t)) \end{aligned} \quad (10)$$

Then, the gain error ε_T can be estimated as

$$\varepsilon_T = \frac{I(t)\tilde{I}(t) - Q(t)\tilde{Q}(t) - (I^2(t) - Q^2(t))}{I^2(t) + Q^2(t)}, \quad (11)$$

Similarly, multiply both sides of (6) by Q and that of (7) by I as

$$\tilde{I}(t)Q(t) \approx I(t)Q(t) + \varepsilon I(t)Q(t) - Q^2(t) \sin \varphi \quad (12)$$

$$\tilde{Q}(t)I(t) \approx Q(t)I(t) - \varepsilon Q(t)I(t) - I^2(t) \sin \varphi \quad (13)$$

Adding (12) and (13) yields:

$$\begin{aligned} \tilde{I}(t)Q(t) + I(t)\tilde{Q}(t) \\ \approx 2I(t)Q(t) - (I^2(t) + Q^2(t)) \sin \varphi \end{aligned} \quad (14)$$

Then, the phase error φ_T can be estimated as

$$\varphi_T = \sin^{-1} \left(\frac{2I(t)Q(t) - (\tilde{I}(t)Q(t) + I(t)\tilde{Q}(t))}{I^2(t) + Q^2(t)} \right). \quad (15)$$

B. Deep learning-based predistortion of I/Q imbalance

Under ideal conditions, the RF feedback-based scheme achieves near-ideal BER performance. However, the performance degradation of predistortion can be observed in environments with noise in the internal circuitry. This performance degradation is considered to result from the process of converting the RF signal through the DAC and the subsequent internal feedback loop involved in the RF feedback-based predistortion scheme. In this subsection, the deep learning-based predistortion of I/Q imbalance is proposed, which does not involve the internal feedback or DAC processes.

In this paper, we employ BLSTM as a deep learning-based approach for I/Q imbalance predistortion. Unlike traditional LSTM, BLSTM processes information bidirectionally, utilizing both past and future context [13]–[15].

At the input layer, the real and imaginary parts of the complex signal affected by I/Q imbalance are separately fed into the BLSTM model. These signals pass through both forward and backward BLSTM layers, which learn time-dependent features. The forward and backward hidden states, denoted by $\vec{h}_t$ and $\overleftarrow{h}_t$, are expressed as

$$\vec{h}_t = LSTM_{forward}(x_t, \vec{h}_{t-1}), \quad (16)$$

$$\overleftarrow{h}_t = LSTM_{backward}(x_t, \overleftarrow{h}_{t+1}). \quad (17)$$

Then, the combined hidden state h_t , obtained by combining the forward and backward hidden states, can be expressed as

$$h_t = [\vec{h}_t, \overleftarrow{h}_t]. \quad (18)$$

The BLSTM model is trained using pairs of signals with I/Q imbalance and their corresponding ideal signals. The loss function L is defined as the Mean Squared Error (MSE) and can be expressed as

$$L = \frac{1}{N} \sum_{i=1}^N (|I_{i,ideal} - I_{i,comp}|^2 + |Q_{i,ideal} - Q_{i,comp}|^2), \quad (19)$$

where $I_{i,ideal}$ and $Q_{i,ideal}$ are in-phase and quadrature part of the ideally predistorted signal, respectively. On the other hand, $I_{i,comp}$ and $Q_{i,comp}$ are in-phase and quadrature part of the predistorted signal obtained through training, respectively. Finally, the model is trained using the training dataset, which consists of OFDM symbols with I/Q imbalance, and the defined loss function based on the MSE. Through the optimization process, the model learns to effectively compensate for I/Q imbalance by minimizing the defined loss between the ideal and the compensated signals.

For I/Q imbalance predistortion, the use of BLSTM algorithm offers several advantages. It can effectively compensate for I/Q imbalance exhibiting non-linear characteristics, which is challenging for conventional linear compensation algorithms. Moreover, BLSTM is capable of learning sequential signal characteristics associated with I/Q imbalance, making it suitable for dynamically changing environments. However, the BLSTM-based I/Q imbalance predistortion also presents certain limitations. It requires a large training dataset to achieve reliable performance, and the training process can be computationally intensive and time-consuming. In addition, the implementation may demand more computational resources compared to traditional compensation techniques.

In this paper, considering the system complexity and hardware limitations, the OFDM system is designed with an FFT size of 32. The proposed BLSTM model consists of an input layer, two BLSTM hidden layers, and an output layer. The first and second BLSTM hidden layers contain 70 and 90 units, respectively, chosen to balance model accuracy and

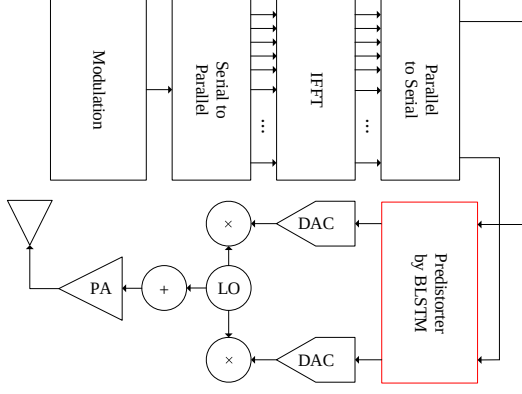


Fig. 2. Transmitter structure with I/Q imbalance predistorter by BLSTM

computational complexity under real-time processing constraints. The first layer is intended to capture fundamental I/Q imbalance patterns across 32 subcarriers, while the second layer is intended to model nonlinear interactions between subcarriers. This achieves stable MSE performance while maintaining computational complexity suitable for practical OFDM processing. To train the BLSTM model, the Root Mean Square Propagation (RMSProp) optimizer was used. The batch size was set to 32, and training was conducted over 60 epochs. The training dataset consisted of a total of 3,200,000 OFDM symbols, which were split into training (2,560,000 symbols) and test (640,000 symbols) sets at a ratio of 4:1.

Fig. 2 shows the OFDM transmitter structure incorporating the proposed BLSTM-based predistorter for I/Q imbalance compensation. In Fig. 2, the BLSTM is applied after the Parallel-to-Serial (P/S) conversion. During the BLSTM processing, the OFDM symbols are predistorted to mitigate the I/Q imbalance and then up-converted to the RF signal.

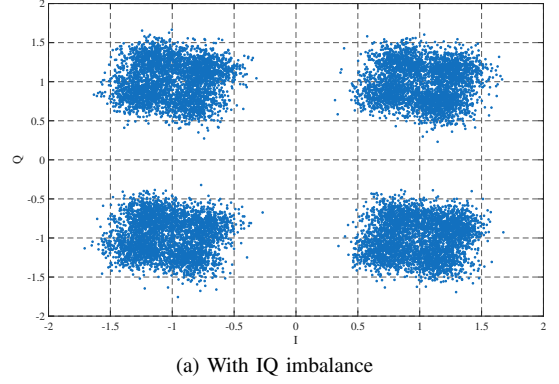
IV. SIMULATION RESULTS

In this section, simulation results are presented. The OFDM signal is modulated by Quadrature Phase Shift Keying (QPSK), and the number of subcarriers is set to 32. For BLSTM training, the input and target data are set to signals with I/Q imbalance and the corresponding ideal signals, respectively. The BLSTM training is performed independently for the I and Q components. The I/Q imbalance is modeled by assuming a gain error uniformly distributed in the range of $0.3 \sim 0.6$ and a phase error uniformly distributed in the range of $5^\circ \sim 8^\circ$.

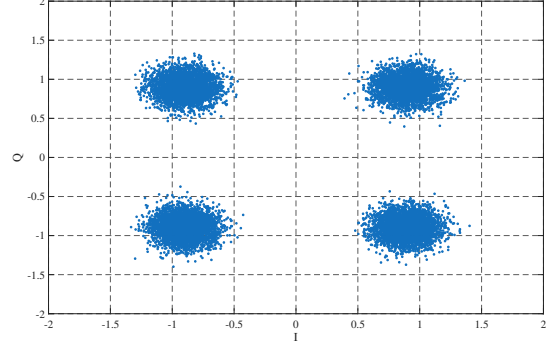
The performance of the training model was evaluated using the MSE. On the test set, the proposed BLSTM-based predistortion model achieved an MSE of 1.2×10^{-4} , demonstrating stable compensation performance under the considered I/Q imbalance conditions. In particular, the model demonstrated stable predistortion performance for I/Q imbalance parameters beyond those explicitly seen during training, indicating its generalization capability.

A. Signal constellation under I/Q imbalance

Fig. 3 shows signal constellations in AWGN at $E_b/N_0 = 15$ dB. In Fig. 3(a), I/Q imbalance causes constellation dispersion,



(a) With IQ imbalance



(b) After BLSTM predistortion

Fig. 3. Signal constellations in AWGN at $E_b/N_0 = 15$ dB.

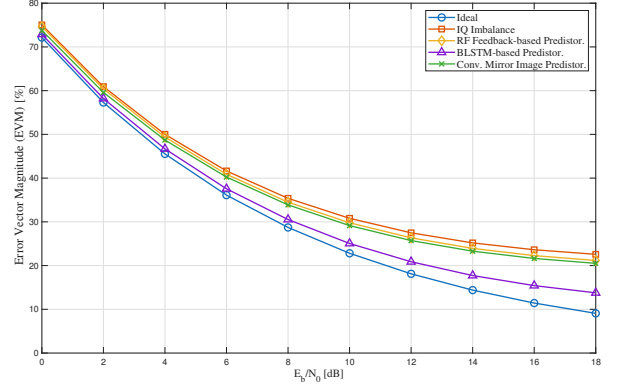


Fig. 4. EVM performance comparison under I/Q imbalance in an AWGN channel.

rotation, and multiple distinct clusters. This characteristic pattern arises because I/Q imbalance is modeled using discrete amplitude and phase error parameters, unlike random RF impairments such as phase noise. In Fig. 3(b), after BLSTM predistortion, the constellation cloud is significantly reduced, demonstrating effective compensation.

B. Error Vector Magnitude (EVM) performance

Fig. 4 shows the Error Vector Magnitude (EVM) performance of the two proposed predistortion schemes in an AWGN channel. For comparison, the results for the uncompensated case and the ideal case (no I/Q imbalance) are also plotted.

At an EVM of 30%, the RF feedback-based scheme achieves approximately 0.5 dB improvement over the uncompensated case, with performance comparable to the conventional mirror image method. At the same EVM of 30%,

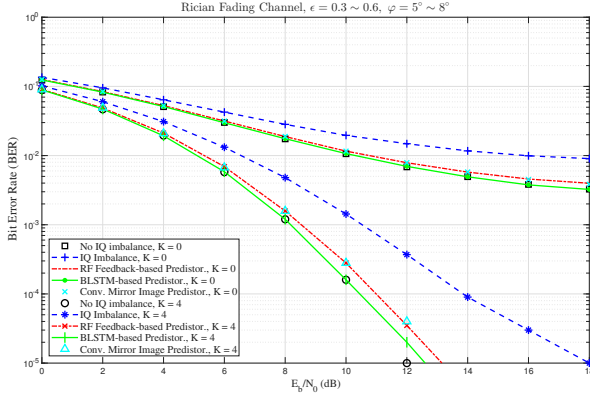


Fig. 5. BER performance with and without I/Q imbalance predistortion over a Rician fading channel.

the BLSTM-based method achieves approximately 2.5 dB improvement over the uncompensated case, and about 2.0 dB improvement over both the RF feedback-based scheme and the conventional mirror image method.

C. BER performance

Fig. 5 shows the BER performance of the two proposed predistortion schemes in a Rician channel with K factors of 0 and 4. For the RF feedback-based scheme, the internal thermal noise in the transmitter feedback path is modeled with an equivalent E_b/N_0 of 15 dB to account for practical hardware imperfections. For comparison, the results for the uncompensated case and the ideal case (no I/Q imbalance) are also plotted. For $K = 0$ at BER of 10^{-2} , and $K = 4$ at BER of 10^{-5} , the RF feedback-based and BLSTM-based schemes achieve approximately 5 dB and 5.8 dB improvements over the uncompensated case, respectively. While the RF feedback-based scheme shows comparable performance to the conventional mirror image method [12], the BLSTM-based scheme further outperforms [12] by 0.8 dB under both channel conditions.

These experimental results confirm that the proposed BLSTM architecture is effective for I/Q imbalance predistortion and achieves performance suitable for practical system implementation under the considered conditions.

V. CONCLUSION

This paper proposed two novel predistortion schemes for I/Q imbalance mitigation in OFDM systems. These two predistortion schemes overcome the pilot-signal and parameter-estimation requirements of conventional approaches. Unlike [9]–[11], they require no pilot signals, and unlike [12], they require no a priori knowledge of I/Q imbalance parameters, eliminating the need for explicit parameter estimation in practice.

Simulation results show that the RF feedback-based predistortion scheme achieves performance nearly identical to the ideal case without I/Q imbalance, whereas the BLSTM-based predistortion scheme shows a performance degradation of about 0.5 dB. The remaining 0.5 dB gap mainly stems from limited training dataset coverage and model capacity constraints. With further improvements in training strategies

and model design, the performance gap between the proposed BLSTM-based predistortion scheme and the RF feedback-based scheme may be further reduced.

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